

THE REMONTADA PROTOCOL

A TACTICAL ECOSYSTEM

Engineering the Comeback

A 3-Part Offline-First Tactical Analytics Architecture

Football matches are lost in the margins: a missed tactical shift at half-time, a player losing focus in the 70th minute, or a miscalculated pass. The Remontada Protocol is an end-to-end tactical ecosystem designed to address three coaching blind spots: network-congested live data capture, momentum-blind spatial models, and result-biased cognitive profiling.

PHASE 1 TAG. DETECT. REPORT. Remontada Command Center	PHASE 2 TRACK. PROJECT. EXPLOIT. Ghost Ball EV	PHASE 3 LOCK. TRACK. ANALYZE. AI Psychiatrist
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SYSTEMS OVERVIEW

This document consolidates the architecture and engineering logic for all three projects. Each phase begins on a separate page, and important formulas are placed on dedicated lines for easier reading.

PHASE 1 | REMONTADA COMMAND CENTER

Touchline Telemetry Suite

TAG. DETECT. REPORT. - Offline-First Distributed State Machine & Heuristic Scouting Engine

A browser-native football match analysis and talent-scouting system designed for the sidelines. The target deployment model is offline-first: no backend, no API keys, and no build step for the browser suite.

Master System Architecture Flow

01	Spatial input - index.html	Touch/click is converted to pitch-container coordinates using the pitch bounding rectangle.
02	Goal-zone check	Normalized coordinates are tested against goal-mouth regions; a qualifying event is tagged as a shot/goal.
03	Cross-context IPC bus	The event payload is serialized as JSON in localStorage under remontada_live_tap.
04	Touchline console - commad-center.html	A storage listener receives pending coordinates and match time; voice input can independently identify a jersey number and action.
05	Event fusion and persistence	fuseEvent() updates an in-memory LIFO event stack and persists match data under scoutai_matches.
06	Undo and video bookmarking	Recent events can be rolled back, while a 10-event window is available for film review.
07	RAE debiasing - bias-detector.html	Raw scout ratings and birth-month multipliers produce an adjusted score and a hidden-gem flag.
08	Dossier engine - report-generatorai.html	Match JSON feeds spatial/temporal heuristics, a 5-axis SVG radar, conditional written verdicts, and A4 PDF export.

Key Phase 1 Formulas

Coordinate normalization	$X = \text{round}(\text{clientX} - \text{rect.left})$ $Y = \text{round}(\text{clientY} - \text{rect.top})$
Goal-zone check	$0.36 \leq Y/H \leq 0.64$ $X \leq 0.07W$ OR $X \geq 0.93W$
Relative Age Effect adjustment	$R_{\text{adj}} = \min(100, \text{round}(R_{\text{raw}} / \beta_m))$
	β_m is a birth-month representation multiplier; the reference design describes a range from 1.40 (January) to 0.70 (December).
Hidden-gem heuristic	$R_{\text{raw}} < 70$ AND $R_{\text{adj}} > 85$

Voice interaction is activated with a hold longer than 350 ms and uses Web Speech API recognition with a constrained JSGF-style grammar for jersey numbers and action verbs.

PHASE 1 | ENGINEERING DETAILS

Deep-Dive Engineering Logic

Cross-context IPC and spatial normalization: The pitch tab and command-center tab are decoupled using `Window.localStorage` as an asynchronous inter-process communication bus. Coordinate plotting does not need to wait for roster selection.

Responsive coordinate mapping: Touch vectors are normalized relative to the pitch container (nominal reference layout 900 × 550 px) and checked against goal-mouth hitboxes for automatic goal-event detection.

Hold-to-speak voice NLP: A wrapper around the Web Speech API constrains recognition to jersey-number tokens and action verbs. A phonetic sanitizer addresses common stadium-noise homophones, for example “for” being normalized to “four”.

Dual-state persistence and deterministic LIFO: Fused events are appended to `masterMatchData` in memory and committed to `localStorage` under `scoutai_matches`. The LIFO stack supports undo, while a 10-event window flags build-up sequences for review.

Relative Age Effect debiasing: Raw performance scores are adjusted by month-based multipliers to reduce the risk of judging youth prospects purely by physical maturity at assessment time.

SVG radar and PDF output: Match JSON is parsed into polar-to-Cartesian coordinates for inline SVG radar charts, then the DOM is captured using `html2canvas` and `jsPDF` for client-side A4 export.

Implementation Reference

Module / mechanism	Responsibility
index.html	Captures pitch touches, normalizes positions, and checks goal-mouth hitboxes.
localStorage IPC	Sends <code>remontada_live_tap</code> payloads between browser contexts via the storage event.
commad-center.html	Combines voice or touch actions with pending position data; persists match events.
bias-detector.html	Applies the birth-month multiplier to scout ratings and flags possible undervalued prospects.
report-generatorai.html	Computes spatial and temporal indicators, generates a five-axis radar, verdict, and downloadable PDF.

Technical interpretation note: the numerical RAE multipliers and hidden-gem thresholds shown here are the values in the supplied architecture description. They should be treated as configurable heuristics and validated against the intended scouting population before use.

PHASE 2 | GHOST BALL EV

Vectorized Spatial Physics Engine

TRACK. PROJECT. EXPLOIT. - Pitch Control, Ray-Cast Passing Interference & Biomechanical Kinematics

Ghost Ball EV evaluates a 104 × 68 pitch grid - 7,072 cells - in a vectorized NumPy pipeline. The intended output is an off-ball expected-value field that highlights accessible spaces, defensive momentum traps, and candidate passing targets in real time.

Master Spatial-Physics Architecture Flow

01	Broadcast video input	Process a broadcast frame (reference size 1280 × 720) with YOLOv8m and ByteTrack for person and ball detections.
02	Pitch-coordinate mapping	Apply perspective homography to map player foot points from image coordinates (u, v) into pitch coordinates (X, Y).
03	Team identification	Extract upper-torso HSV features and cluster teams with K-Means (k = 2), reinforced by a 30-frame temporal vote.
04	Kinematics	Estimate motion using central differences, clamp player speed, and interpolate the ball trajectory when needed.
05	Vectorized xT and offside mask	Evaluate a non-linear expected-threat grid and mask cells beyond the live second-last-defender line.
06	Time-to-intercept model	Combine reaction time, travel distance, speed and turning-orientation penalty to estimate player arrival time.
07	Arrival probability and ray-cast	Convert arrival-time differences into probability waves and apply Gaussian shadowing for defender interference along passing rays.
08	Pressure, smoothing and overlay	Account for local carrier pressure and safe-pass boost, smooth the field with EMA, then render the Ghost Ball or a blocked-lane tripwire.

Core Spatial Formulas

Pitch grid	$104 \times 68 = 7,072 \text{ cells}$
Pitch coordinates	$X \in [0, 104] \text{ m}$ $Y \in [0, 68] \text{ m}$
Speed clamp	$v_{\text{max}} \leq 10.0 \text{ m/s}$
Time to intercept	$t_{\text{player}} = t_{\text{react}} + (d / v_{\text{max}}) + \tau_{\text{turn}}(\cos \theta)$
Logistic arrival probability	$P_i(g) = 1 / (1 + \exp(-3.2 \times (t_{\text{ball}} - t_{\text{player}})))$

PHASE 2 | ENGINEERING DETAILS

Deep-Dive Engineering Logic

Planar homography and kinematics: A 3×3 homography matrix H translates broadcast pixels into metric pitch coordinates. Ground contact is anchored to the bottom-centre/feet position of the player bounding box. HSV K-Means team clustering is stabilized with a 30-frame temporal vote.

Dynamic expected threat and offside masking: The xT grid is bounded dynamically by the X-coordinate of the second-last defender, creating a tactical mask that changes with the defensive line.

Biomechanical turnaround penalty: A defender moving away from a target incurs extra estimated time to decelerate and pivot. The penalty is based on the angle between current velocity and the direction to the target.

Vectorized ray-cast interference: The pipeline casts a ray from the ball to each grid cell and projects all defenders against those rays in vectorized form. Gaussian lane shadowing reduces arrival probability based on perpendicular distance to a passing lane.

EMA filtering and laser tripwire: A 10-frame exponential moving average stabilizes the EV grid. The highest-EV coordinate is projected as a Ghost Ball unless a defender intersects the lane within the stated interception distance.

Formula Reference

Orientation cosine	$\cos(\theta) = (v \cdot d) / (v \times d)$
	v is the defender velocity vector; d points from the defender toward the target location.
Ray-cast tripwire condition	$0.08 < t < 0.95$ $d_{ray} < 1.15 \text{ m}$
	The lane is considered blocked when the projected intersection falls within the specified time range and distance threshold.
Local carrier pressure radius	$R \leq 5.0 \text{ m}$
Exponential moving average	$\alpha \approx 0.18$ Window = 10 frames

The overlay uses inverse homography (H^{-1}) to project the recommended location back onto the video frame. A blocked lane is shown with a crimson ray and a “BLOCKED” shatter effect; a clear lane displays the pulsing Ghost Ball and dominance ripples.

PHASE 3 | AI PSYCHIATRIST

Cognitive Profiling & Computer Vision

LOCK. TRACK. ANALYZE. - Measuring La Pausa, Tunnel Vision and Cognitive Decay

The AI Psychiatrist Engine maps broadcast-video observations into a cognitive and tactical profile. It aims to quantify mental-fatigue proxies, spatial-scanning frequency, and likely sources of tactical failure without wearable GPS trackers.

Master Cognitive Pipeline Flow

01	Target acquisition	The operator selects a target region with cv2.selectROI(); the bounding box and name are saved to JSON.
02	Ball tracking	A YOLO class-32 scanner detects the ball. When the ball is temporarily lost, a memory buffer interpolates for up to 30 frames.
03	Telemetry and BotSORT tracking	Target ROIs are associated with detections using an IoU matching threshold; a five-frame macro-heading filter smooths direction estimates.
04	Scanning estimate	Circular Standard Deviation is calculated from heading estimates; values above the configured threshold indicate a potential scanning event.
05	Possession state buffer	A 10-frame possession-state buffer maintains timing continuity when the ball is occluded.
06	Cognitive and tactical models	Cognitive decay, accidental-genius vector alignment, tactical space value and adjusted velocity (V_adj) are evaluated.
07	Bayesian fault isolation	The model estimates posterior responsibility/failure likelihood from evidence and prior structural-failure likelihoods.
08	HUD rendering	Render vision cones, 0-100% focus bars, and “La Pausa” versus “Hesitation” labels on the video.

Core Cognitive Formulas

Target association	$\text{IoU} > 0.20$
Macro-heading lookback	Lookback = 5 frames
Scanning threshold	$\sigma_{\text{heading}} > 10.0^\circ$
Cognitive decay	$C(t_d) = e^{(-k \times t_d)}$
Accidental Genius alignment	$\text{angle}(u, v) \leq 10.0 \text{ degrees}$
	The initial scan direction u is compared with the final pass direction v.

PHASE 3 | ENGINEERING DETAILS

Deep-Dive Engineering Logic

BotSORT target handshake: Operator-selected regions of interest are associated with YOLO bounding boxes using an Intersection over Union threshold greater than 0.20.

Circular Standard Deviation for spatial scanning: A five-frame macro-heading filter estimates net displacement so small tracking jitter is less likely to be interpreted as scanning. Circular statistics use the mean cosine and sine of headings to handle wraparound at 360 degrees.

10-frame possession buffer: The buffer limits timer resets when the ball is briefly occluded, for example behind a referee. The processing timer is intended to continue through short detection gaps so La Pausa metrics are not erased by visual noise.

Cognitive mathematical model: Focus degradation under pressure is represented by an exponential decay model. The “Accidental Genius” check compares the initial scan direction with the final pass vector to test whether the action was directionally anticipated.

Bayesian fault isolation: Bayes’ theorem is used to distribute likely responsibility across the front three according to evidence and structural failure priors, reducing dependence on the final outcome alone.

Formula Reference

Bayes’ theorem	$P(A \mid E) = [P(E \mid A) \times P(A)] / P(E)$
	A is a hypothesized cause or failure assignment; E is the observed evidence.
Cognitive decay over elapsed duration	$C(t_d) = e^{(-k \times t_d)}$
Focus indicator range	$0 \leq \text{Focus} \leq 100\%$
Hesitation trigger	$\text{Focus} < 65\%$
La Pausa indicators	Latency > 1.5 s AND Space value increases
Scanning estimate	$\sigma_{\text{heading}} > 10.0^\circ$
	The source design describes this as a potential shoulder-check/spatial-scanning indicator, not a direct measurement of internal mental state.

Interpretation caution

These outputs are computer-vision-derived proxies and heuristics. Broadcast footage alone cannot directly measure a player’s mental state, fatigue, intent, or attention. Validate thresholds on representative clips and treat cognitive labels as analytical hypotheses rather than clinical diagnoses or ground truth.

END OF MASTER ARCHITECTURE DOCUMENT